Scaling ANN-to-SNN Conversion to Large Language Models via Representation and Local Approximation

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Abstract

ANN-to-SNN conversion has struggled at large language model (LLM) scale, where prior methods exhibit severe degradation due to error accumulation across deep transformer stacks. We identify the root cause as structural: existing approaches treat conversion as a layer-wise approximation problem, causing errors to compound with depth. We instead decompose conversion into representation and approximation. Bit Spike Encoding (BSE) provides a bijective mapping between quantized values and spike trains, enabling linear layers to be computed exactly relative to their quantized ANN counterparts, without introducing additional conversion error. This reduces the problem to nonlinearities, which are the only remaining approximation sites. We address these with Adaptive Spiking Neural Codec (ASNC), a distribution-aware piecewise scheme that produces bounded approximation error and re-encodes outputs into the BSE format, thereby localizing error at each nonlinear module. Under this formulation, we derive an end-to-end error bound showing that conversion error is governed by nonlinear modules and their Lipschitz properties, rather than by network depth. Empirically, LASER converts LLMs up to 70B parameters under an FP16/16-step regime with less 2% degradation across MMLU, HellaSwag, ARC, and TruthfulQA, without degradation scaling across models from 32 to 80 transformer blocks.

1 Introduction

Large language models are exactly the workload where neuromorphic hardware would pay off most: billion-parameter transformers dominate inference energy budgets, and event-driven substrates like Intel Loihi 2 offer order-of-magnitude energy reductions if the model can be expressed as an SNN [Davies et al., 2018, 2021]. Yet ANN-to-SNN conversion—the natural route, since it preserves the capability learned during pretraining—has not crossed the threshold to LLM scale. Prior conversion methods report perplexity degradations in the double digits on billion-parameter transformers, or require timestep budgets of 10^3 or more to approach acceptable accuracy [Xing et al., 2025, Zhou et al., 2024, Zhao et al., 2025]. Conversion for LLMs has been, in practical terms, an unsolved problem.

The mechanism is structural. Every existing conversion path introduces approximation *at every layer*: lossy spike encodings (rate coding, TTFS) at every activation, architectural substitution at every nonlinearity [Zhou et al., 2023, Yao et al., 2023, Hwang et al., 2024], and heuristic surrogate gradients at every backward step [Neftci et al., 2019, Shrestha and Orchard, 2018, Deng et al., 2022]. In a shallow vision network these per-layer errors are manageable. In a transformer with tens of layers they begin to accumulate; in a 32-block LLM they compound to the point of failure; in a 70B-parameter 80-block model they saturate any reasonable accuracy budget. The problem is not any individual conversion step, but the accumulation of error through depth, which LLMs are too deep to absorb.

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The right place to intervene is the representation itself. Prior methods treat conversion as an *approximation* problem: continuous values are approximated by discrete spike trains at every layer, and the resulting error accumulates through depth. While each approximation may be small, its repeated application leads to large degradation in deep models such as LLMs. The issue is therefore not any individual conversion step, but that approximation is introduced everywhere, causing total error to grow with model depth. To address this, we reduce both the amount and the spread of approximation. An N-bit quantized value already admits an exact discrete representation—its binary expansion—and a spike train of length N can encode it without loss. This allows linear layers to be implemented without introducing additional conversion error, reducing the linear path to standard quantized ANN computation. The remaining approximation is confined to nonlinear operations, which cannot be made exact in the discrete domain.

We realize this decomposition in LASER through two components. **Bit Spike Encoding (BSE)** provides a bijective mapping between N-bit quantized values and N-timestep spike trains, ensuring that linear layers match their quantized ANN counterparts. **Adaptive Spiking Neural Codec (ASNC)** handles nonlinearities with bounded, distribution-aware approximation and re-encodes outputs into the same representation, so that approximation error remains localized. As a result, total conversion error depends only on a small set of nonlinear modules, rather than accumulating across all layers, enabling stable conversion even in deep LLMs. Our main contribution are:

- **Structural diagnosis.** We identify depth-wise error accumulation as the key limitation of prior ANN-to-SNN conversion methods at LLM scale, arising from approximation introduced at every layer.
- **Representation–approximation decomposition.** We propose a framework that eliminates approximation in linear layers via Bit Spike Encoding (BSE) and confines it to nonlinear operations via Adaptive Spiking Neural Codec (ASNC), reducing both the amount and spread of error.
- **End-to-end error analysis.** We show that, under this decomposition, total conversion error depends only on nonlinear modules and their Lipschitz properties, rather than accumulating across all layers.
- **LLM-scale empirical validation.** We demonstrate stable conversion of transformers up to 70B parameters under a 16-step regime, with minimal degradation and no systematic growth with model depth.

2 Related Work

ANN-to-SNN conversion has been studied along three axes, each contributing its own approximation errors.

Spike encoding. Rate coding [Rueckauer et al., 2017, Auge et al., 2021, Guo et al., 2021] requires thousands of timesteps for acceptable precision; TTFS [Bonilla et al., 2022, Stanojevic et al., 2024] reduces latency at the cost of a single-spike constraint, and TTFS-Former still requires 1,024 steps for FP16 precision [Zhao et al., 2025]. Spikformer and Spikformer-v2 [Zhou et al., 2023, 2024] and SpikeZIP-TF [You et al., 2024] trade latency for accuracy via long time windows. All these schemes are lossy many-to-one mappings between continuous values and spike trains. BSE differs fundamentally in being bijective: it represents an N-bit integer by its binary expansion across N timesteps, so no information is lost in transit.

Nonlinear computation. SiLU, Softmax, and LayerNorm have no exact spike-domain equivalent, forcing either structural substitution—SSA in Spikformer [Zhou et al., 2023], rewrites in Spike-Driven Transformer [Yao et al., 2023, Hwang et al., 2024, Wang et al., 2023]—or approximate equivalence [You et al., 2024, Tang et al., 2025, Zhu et al., 2024, Xing et al., 2025]. Both paths inject per-site approximation errors. ASNC differs from prior approximate approaches in two ways: it allocates segments non-uniformly to match the empirical activation density, and its outputs are re-encoded into BSE format so that downstream linear layers inherit the BSE–quantized-ANN equivalence.

Training. Surrogate gradients [Neftci et al., 2019, Wu et al., 2018, Shrestha and Orchard, 2018] are the standard remedy for non-differentiable spikes, applied globally without any guarantee of forward-backward consistency. STBP [Wu et al., 2018], SLAYER [Shrestha and Orchard, 2018], and

DIET-SNN [Rathi and Roy, 2023] reduce stability at scale; conversion with fine-tuning [Rathi et al., 2020] and direct training [Xing et al., 2024] share the same limitation. Our STE characterization under LASER is narrower in scope: we do not claim an unconditional surrogate-gradient remedy, only that under BSE and ASNC the forward-backward mismatch can be characterized layer-type by layer-type rather than treated as a global heuristic.

3 Preliminaries

SNN Computation Model. We adopt the non-leaky Integrate-and-Fire (IF) model with soft reset and dynamic threshold as the computational primitive. Given input current $I_{\text{in}}(t)$, membrane potential $V_m(t)$, and dynamic threshold $\Theta(t)$, the neuron emits spike $S(t) \in \{0, 1\}$ and updates via

$$S(t) = \mathbb{1}[V_m(t) + I_{\text{in}}(t) \geq \Theta(t)]\,, \qquad V_m(t{+}1) = V_m(t) + I_{\text{in}}(t) - S(t)\,\Theta(t), \tag{1}$$

initialized with $V_m(0) = 0$. For a layer of d neurons, these updates apply elementwise, yielding $\mathbf{S}(t) \in \{0, 1\}^d$. A spiking layer maps an input spike sequence to an output spike sequence via these dynamics, and stacking such layers defines the network function $\mathcal{F}_{\text{SNN}}$. Setting $\Theta(t)$ to a fixed positional schedule realizes BSE; making it learnable realizes ASNC. A layer's output is the spike train $\mathbf{S} = (S(0), \ldots, S(T-1)) \in \{0, 1\}^T$.

Formal Problem Statement. Let $\mathcal{F}_{\text{ANN}}$ be a feedforward network alternating affine operations $\mathbf{z}^\ell = \mathbf{W}^\ell \mathbf{x}^{\ell-1} + \mathbf{b}^\ell$ and nonlinearities $\mathbf{x}^\ell = \sigma(\mathbf{z}^\ell)$. Let $\mathcal{F}_{\text{QANN}}$ denote its N-bit quantized counterpart. ANN-to-SNN conversion seeks a spiking network $\mathcal{F}_{\text{SNN}}$ such that

$$\left\|\mathcal{F}_{\text{SNN}}(\mathbf{x}^0) - \mathcal{F}_{\text{QANN}}(\mathbf{x}^0)\right\| \;\leq\; \epsilon_{\text{conv}}, \tag{2}$$

with all inter-layer communication realized via binary spike trains. The total deviation from the full-precision ANN decomposes as

$$\|\mathcal{F}_{\text{SNN}}(\mathbf{x}^0) - \mathcal{F}_{\text{ANN}}(\mathbf{x}^0)\| \;\leq\; \epsilon_{\text{conv}} + \epsilon_{\text{quant}},$$

where ϵ_{quant} is the quantization error.

Worst-case conversion-error accumulation. Let $\mathbf{y} = \mathcal{F}_{\text{QANN}}(\mathbf{x}^0)$ denote the N-bit quantized ANN output and $\hat{\mathbf{y}} = \mathcal{F}_{\text{SNN}}(\mathbf{x}^0)$ its SNN counterpart. Suppose layer ℓ introduces relative conversion error ϵ_ℓ, and the nonlinearities are L_σ-Lipschitz. Then

$$\|\hat{\mathbf{y}} - \mathbf{y}\| \;\leq\; \|\mathbf{y}\| \prod_{\ell=1}^{L} (1 + L_\sigma \epsilon_\ell) \;=\; O\big((1+\epsilon)^L\big)\,, \tag{3}$$

where $\epsilon = \max_\ell \epsilon_\ell$. This bound motivates LASER: if conversion error is eliminated for linear layers and bounded only at nonlinear layers, then accumulation is governed by the nonlinear modules rather than by all layers in the network. The total deviation from the full-precision ANN further decomposes into conversion error plus the fixed quantization floor.

4 Methodology

LASER converts a pretrained ANN to an SNN by directly controlling the conversion error relative to the quantized ANN. The design is motivated by the accumulation bound in (3): if every layer introduces approximation error, the deviation compounds geometrically in depth. LASER eliminates this source of accumulation by making linear layers lossless relative to the quantized ANN and confining all approximation to nonlinear modules, where the error is explicitly bounded.

LASER has two architectural components. **BSE** (Section 4.1) is a parameter-free encoding that makes linear layers lossless relative to the quantized ANN. **ASNC** (Section 4.2) is a learnable piecewise codec that bounds the error at nonlinear modules. Because ASNC has learnable parameters and contains non-differentiable operators, its specification comprises three parts: the codec itself, a forward-pass error analysis, and a training procedure with a matching backward-pass analysis. Together these yield a unified depth-independence result: errors accumulate only at nonlinear sites in *both* the forward pass (Theorem 4.10) and the backward pass (Proposition 4.14).

Operators used throughout. We fix notation for three operators that appear repeatedly.

(i) *N-bit quantizer.* For calibration (λ, μ), $\mathcal{Q}_N^{(\lambda,\mu)} : \mathbb{R} \to \{0, 1, \ldots, 2^N - 1\}$,

$$\mathcal{Q}_N^{(\lambda,\mu)}(x) \;=\; \mathrm{clip}\big(\lfloor \lambda(x+\mu) \rceil,\; 0,\; 2^N - 1\big),$$

with rounding-to-nearest and saturation at the grid endpoints. When (λ, μ) is clear from context we write $\mathcal{Q}_N$.

(ii) *IF spike-emission operator.* For a threshold schedule $\mathbf{\Theta} = (\Theta(0), \ldots, \Theta(T-1)) \in \mathbb{R}^T_{>0}$ and input current sequence $\mathbf{u} \in \mathbb{R}^T$,

$$\Phi_T^{\mathrm{IF}}\big(\mathbf{u};\, \mathbf{\Theta}\big) \;:\; \mathbb{R}^T \longrightarrow \{0,1\}^T$$

denotes the length-T spike train produced by iterating the dynamics of (1) with $V_m(0) = 0$ and $I_{\mathrm{in}}(t) = u(t)$. Impulse input is the special case $\mathbf{u} = q\,\mathbf{e}_0$, with $\mathbf{e}_0$ the first standard basis vector.

(iii) *BSE encoder and decoder.* $\mathsf{E}_N(q) \;=\; \Phi_N^{\mathrm{IF}}\big(q\,\mathbf{e}_0;\, \mathbf{\Theta}_{\mathrm{BSE}}\big)$ under the positional schedule $\mathbf{\Theta}_{\mathrm{BSE}}(t) = 2^{N-1-t}$, and $\mathsf{D}_N(\mathbf{S}) \;=\; \langle \mathbf{S},\, \mathbf{\Theta}_{\mathrm{BSE}} \rangle$.

BSE and ASNC thus share a single computational primitive, the IF operator Φ_T^{IF}, under two different threshold schedules: a fixed positional schedule for BSE and a learned schedule for ASNC.

4.1 Bit Spike Encoding (BSE): Linear Layers at the Quantization Floor

Existing spike codes incur irreducible error on top of quantization: rate coding accumulates counts over a time window, losing precision at every step; TTFS maps values to timing, which is inherently imprecise at finite resolution. We ask whether an N-bit quantized value can be represented as a spike train with *zero additional loss*. The binary numeral system answers yes, and BSE realizes this representation via a single IF neuron driven by an impulse. BSE is parameter-free: its threshold schedule and calibration are fixed by the target ANN's quantization grid.

Assumption 4.1 (Quantization-grid alignment)**.** For each layer ℓ, BSE uses an N-bit window with per-channel affine calibration (λ_ℓ, μ_ℓ) matched to the ANN's own quantization grid. Activations outside the clipping range $[\mu_\ell,\, \mu_\ell + (2^N - 1)/\lambda_\ell]$ are clipped identically to the ANN. The range parameters (λ_ℓ, μ_ℓ) are estimated from a calibration pass and assumed stable at test time (sensitivity validated in Section 5.4).

An N-bit integer $q = \sum_{t=0}^{N-1} b_t\, 2^{N-1-t}$ is unfolded across time by the BSE encoder and recovered by the decoder:

$$\mathsf{E}_N(q) \;=\; \Phi_N^{\mathrm{IF}}\big(q\,\mathbf{e}_0;\, \mathbf{\Theta}_{\mathrm{BSE}}\big), \qquad \mathbf{\Theta}_{\mathrm{BSE}}(t) = 2^{N-1-t}, \tag{4}$$

$$\mathsf{D}_N(\mathbf{S}) \;=\; \langle \mathbf{S},\, \mathbf{\Theta}_{\mathrm{BSE}} \rangle \;=\; \sum_{t=0}^{N-1} S(t)\, 2^{N-1-t}. \tag{5}$$

Operationally, a spike-encoded activation is treated as an integer on the N-bit grid: weight-activation products reduce to exact integer multiply-accumulate, and the output is re-quantized via $\mathcal{Q}_N$ onto the downstream grid.

Activation–activation products. The product of two BSE spike trains is not itself a valid BSE representation. For operations such as the Query-Key product $\mathbf{q}^\top \mathbf{k}$ we use *decode-compute-re-encode* (DCR): decode both operands via D_N, compute the product with standard MAC hardware, and re-encode via E_N. DCR incurs only the standard N-bit quantization error of the product and preserves Corollary 4.4 on all downstream layers. Cost analysis appears in Appendix A.

Theoretical Analysis. The construction of (4)–(5) has two consequences: it is bit-exact at the neuron level, and functionally identical to the quantized ANN at the layer level.

Lemma 4.2 (Bit-exact emission)**.** *Under schedule $\mathbf{\Theta}_{\mathrm{BSE}}$ and impulse input $q\,\mathbf{e}_0$, the IF operator Φ_N^{IF} emits $S(t) = b_t$ for all $t = 0, \ldots, N-1$. The membrane potential satisfies $V_m(t) = \sum_{\tau > t} b_\tau\, 2^{N-1-\tau}$, tracking the remaining unprocessed bits at each step.*

The proof is by induction on t (Appendix C).

Theorem 4.3 (Bijectivity and entropy preservation)**.** *E_N and D_N are mutual inverses: $\mathsf{D}_N(\mathsf{E}_N(q)) = q$ for all $q \in \mathcal{Q}_N$ and $\mathsf{E}_N(\mathsf{D}_N(\mathbf{S})) = \mathbf{S}$ for all $\mathbf{S} \in \{0,1\}^N$. Consequently $H(\mathsf{E}_N(Q)) = H(Q)$ for any integer-valued Q: BSE preserves Shannon entropy exactly.*

Corollary 4.4 (Linear layers match the quantized ANN forward pass)**.** *Under Assumption 4.1, the BSE forward pass through any linear layer is functionally identical to the corresponding N-bit quantized ANN forward pass. Any discrepancy from full-precision arithmetic equals the ANN's own N-bit quantization error. In terms of* (3)*, $\epsilon_\ell = 0$ for every linear layer relative to the quantized baseline.*

The proof uses the soma accumulation $Q_a = \mathsf{D}_N(\mathsf{E}_N(q_a)) = q_a$, making weight-activation products exact integer arithmetic over the grid (Appendix C).

Scope of Corollary 4.4. The corollary states that BSE introduces no *additional* error at a linear layer: any input error e is passed through as Ae, just as in the quantized ANN. It does *not* say perturbations are removed by linear layers. Two mechanisms keep such perturbations from accumulating: *(i) re-quantization as a firewall.* Each linear output is re-quantized via $\mathcal{Q}_N$ onto the N-bit grid. Measured in per-layer normalized grid units (Definition 4.9), this absorbs $\|A\|$ into the downstream calibration scale, so grid-unit errors do not acquire spectral-norm factors across layers. In absolute activation units a $\prod_m \|A_m\|$ factor reappears; we report these factors empirically in Section 5.4. *(ii) sub-grid error absorption.* Any input error with magnitude below half the grid spacing $\delta_{\text{grid}} = 1/\lambda_\ell$ is rounded out by $\mathcal{Q}_N$; only larger errors survive as integer shifts. At $N = 16$ we empirically observe ASNC output errors well below the grid spacing across layers (Table 5).

4.2 Adaptive Spiking Neural Codec (ASNC): Bounding Nonlinear Error

Corollary 4.4 reduces the conversion-error problem to bounding the error at nonlinear modules. ASNC approximates a target nonlinearity σ (SiLU, Softmax, GELU) by a distribution-aware piecewise codec over K segments and immediately re-encodes its output into BSE format so that the corollary applies to all downstream linear layers. ASNC has learnable parameters, so its specification comprises three parts: the codec construction (Section 4.2), a forward-pass error analysis (Section 4.2), and a training procedure with a matching backward-pass analysis (Section 4.2).

Construction.

Assumption 4.5 (Near-Gaussian activations)**.** The scalar activation inputs to each nonlinear module have a density well-approximated by $\mathcal{N}(\mu_a, \sigma_a^2)$ over the clipping range, consistent with empirical observations in large transformers after LayerNorm [He et al., 2024, Lin et al., 2024]. The empirical validity of this assumption is tested on LLaMA-2 7B in Section 5.4; we also report behavior under deliberate violation of the assumption.

Under Assumption 4.5, segments $\{[a_i, b_i]\}_{i=1}^K$ are allocated non-uniformly: high-density and outlier regions receive finer partitions, low-density regions coarser ones. For segment i, a compact codec with learned parameters $(s_i, c_i, \boldsymbol{\theta}_i, \mathbf{w}_i)$ produces a scalar estimate by applying an affine pre-transform, the N-bit quantizer, the IF operator under a learned threshold schedule, and a learned linear read-out:

$$\mathbf{s}_i(x) \;=\; \Phi_T^{\text{IF}}\Big(\mathcal{Q}_N\big(s_i\, x + c_i\big)\, \mathbf{e}_0;\, \boldsymbol{\theta}_i\Big) \;\in\; \{0,1\}^T, \qquad \mathcal{C}_i(x) \;=\; \langle \mathbf{w}_i,\, \mathbf{s}_i(x)\rangle. \tag{6}$$

A hard segment selector $\mathcal{I} : \mathbb{R} \to \{1, \ldots, K\}$ with indicators $\mathbb{1}_{\mathcal{I}(x)=i}$ satisfying $\sum_{i=1}^K \mathbb{1}_{\mathcal{I}(x)=i} \equiv 1$ routes each input to exactly one segment. The scalar codec output is immediately re-encoded into BSE format:

$$\mathbf{S}^{\text{BSE}}(x) \;=\; \sum_{i=1}^{K} \mathbb{1}_{\mathcal{I}(x)=i}\; \mathsf{E}_N\big(\mathcal{C}_i(x)\big) \;\in\; \{0,1\}^N. \tag{7}$$

The partition-of-unity property of the indicators guarantees that exactly one term in (7) is nonzero per input, so $\mathbf{S}^{\text{BSE}}(x)$ is a valid BSE spike train by construction and Corollary 4.4 applies immediately at the next linear layer. Adaptive segment splitting and morphology-aware weighting are detailed in Appendix F.

Forward-Pass Analysis. We first bound the per-module distortion, then combine with Corollary 4.4 to obtain the end-to-end forward bound.

Theorem 4.6 (ASNC distortion at the Lloyd–Max optimum)**.** *Let σ be Borel-measurable and let D_K^{ASNC} denote the mean-squared distortion of ASNC at convergence. At the Lloyd–Max optimum (per-segment reconstruction points coinciding with the conditional mean of $\sigma(X)$ on each segment), $D_K^{\text{ASNC}} = O(K^{-2})$, with a non-uniform allocation giving a strictly smaller constant than uniform under Assumption 4.5. Proof in Appendix D.*

Remark 4.7 (On codec convergence). Theorem 4.6 characterizes the best achievable distortion given the hypothesis class. Whether the codec (6) trained by gradient descent reaches this optimum is a separate empirical question. The segment-wise universal-approximation property of the IF codec (Lemma D.1 in the appendix) shows that the Lloyd–Max reconstruction points are *representable* by the codec, but does not establish global convergence of gradient descent on the non-convex non-smooth training objective. The empirical scaling of D_K with K under our training procedure is reported in Section 5.2 (Figure 1), where the fitted slope is close to the predicted -2 on log–log axes.

Assumption 4.8 (Lipschitz-bounded nonlinearities)**.** Each nonlinear activation σ_m has Lipschitz constant L_m: $|\sigma_m(a) - \sigma_m(b)| \leq L_m|a - b|$ on the calibrated clipping range. For SiLU on the empirical input range we observe $L_m \lesssim 1.1$; for row-wise Softmax, $L_m \leq 1$ (with the tighter bound $\max|\partial\alpha/\partial s| \leq 1/4$ for attention scores, Appendix A).

To lift the per-module bound to a network-level statement we need a convention that eliminates spurious spectral-norm factors introduced by linear layers.

Definition 4.9 (Normalized grid units)**.** For each layer ℓ with calibration scale λ_ℓ, let $\tilde{\mathbf{x}}^\ell = \lambda_\ell(\mathbf{x}^\ell + \mu_\ell)$ denote the activation in *normalized grid units* (so $\tilde{\mathbf{x}}^\ell$ lies on the integer grid $[0, 2^N - 1]$). Layer-to-layer error $\tilde{e}_\ell$ is measured in these units.

In grid units the linear map A_m between nonlinear modules is a map from one integer grid to the next followed by re-quantization via $\mathcal{Q}_N$. Under Assumption 4.1 this re-quantization absorbs $\|A_m\|$ into the downstream calibration scale $\lambda^{\text{out}}_{m+1}$, so the grid-unit norm of the linear-map output error is bounded by the grid-unit input error plus one rounding. A precise statement is given in Appendix B.

Theorem 4.10 (End-to-end forward error bound under BSE and ASNC)**.** *Under Assumptions 4.1, 4.5, and 4.8, let the network alternate M nonlinear modules (ASNC error $\epsilon_m \leq D_K^{\text{ASNC}}$) and linear layers (BSE, matching the quantized ANN up to a rounding). With $\bar{L} = \max_m L_m$, the total conversion error relative to the N-bit quantized ANN, measured in per-layer normalized grid units, satisfies*

$$\left\|\tilde{\mathbf{y}} - \tilde{\mathbf{y}}_{\text{quant}}\right\| \;\leq\; \left(\sum_{m=1}^{M} \bar{L}^{M-m}\right)\epsilon_{\text{ASNC}} \;+\; O(M \cdot \epsilon_{\text{quant}}), \tag{8}$$

where $\epsilon_{\text{ASNC}} = \max_m \epsilon_m$. In absolute activation units the bound acquires an additional factor $\prod_m \|A_m\|$. Full proof in Appendix B.

Corollary 4.11 (Forward error accumulates only at nonlinear sites)**.** *Under the conditions of Theorem 4.10, the grid-unit bound depends on M (the number of nonlinear modules) and $\bar{L}$, but* not *on the number of linear layers $L - M$. For $\bar{L} \leq 1$ (Softmax), the sum converges to $(1 - \bar{L})^{-1}\epsilon_{\text{ASNC}}$ as $M \to \infty$; for $\bar{L} > 1$ (SiLU, empirically $\bar{L} \approx 1.1$), the sum grows as $(\bar{L}^M - 1)/(\bar{L} - 1)$, geometric in M.*

Remark 4.12 (On "depth-independence"). We use "depth-independent" in the specific sense of Corollary 4.11: the bound does not grow with the number of linear layers between nonlinear modules. It *does* grow with M, which in transformers scales linearly with depth (typically $M = 3L_{\text{blocks}}$ for one SiLU, one Softmax, and one LayerNorm per block). Hence "depth-independent" does not mean "constant in depth." For LLaMA-2 7B with $L_{\text{blocks}} = 32$, $M = 96$ and $\bar{L}^M \approx (1.1)^{96} \approx 1.25 \times 10^4$; combined with an empirical $\epsilon_{\text{ASNC}} \approx 10^{-6}$, the resulting bound product is ≈ 0.013, consistent with the empirical $\Delta\text{PPL} \approx 0.46$. The worst-case bound is conservative; the tighter practical value uses per-module L_m and the fact that errors at different sites partially cancel rather than add at their worst-case signs.

Training and Backward-Pass Analysis. ASNC's learnable parameters $\{(s_i, c_i, \boldsymbol{\theta}_i, \mathbf{w}_i, [a_i, b_i])\}_{i=1}^K$ must be adapted to the target model. We describe the training procedure and then analyze its gradient behavior, showing that the depth-independence property established for the forward pass (Corollary 4.11) extends to the backward pass as well.

Training procedure. ASNC parameters are adapted to a pretrained target model using 1,024 MMLU calibration samples, without modifying the original ANN weights. Because the BSE encoder E_N (appearing on the output side of (7)) and the segment selector $\mathbb{1}_{\mathcal{I}(x)=i}$ are non-differentiable, we apply the *hard Straight-Through Estimator* (STE): the Jacobian of E_N is treated as the identity, $\partial\mathbf{S}_{\text{out}}/\partial V_{\text{in}} = \mathbf{I}$, and the same convention is applied through the piecewise-constant step induced by the segment selector. This is a training-time device only; at inference, the true discrete operators are used.

Why a layer-wise analysis is possible under LASER. Prior SNN work applies surrogate gradients uniformly across all layers without any guarantee of forward–backward consistency, so gradient mismatches cannot be attributed to specific layer types. Under LASER the BSE/ASNC split separates *parameter-free, exact* linear layers from *parameterized, approximated* nonlinear modules; this separation carries over to the backward pass and permits a layer-wise decomposition of the STE mismatch.

Assumption 4.13 (Minimum input spacing within segments)**.** Within each ASNC segment i, the minimum spacing between distinct quantization levels in $[a_i, b_i]$ is $\delta_i > 0$, determined by the affine calibration and the N-bit grid. We write $\delta_{\min} = \min_i \delta_i$.

Proposition 4.14 (STE gradient decomposition under LASER)**.** *Under Assumptions 4.1–4.13:*

1. ***Linear layers.*** *The STE gradient equals the true gradient of the N-bit quantized ANN exactly. No additional mismatch is introduced beyond what the quantized ANN itself incurs under standard QAT-style identity substitution for $\mathcal{Q}_N$. Upstream gradients flowing back to ASNC parameters therefore pass through BSE-coded linear layers without LASER-specific distortion.*

2. ***Nonlinear layers.*** *At an ASNC module, the STE delivers the upstream gradient g as if the module were identity, while the true gradient is $g \cdot \sigma'(x)$. The resulting mismatch has two parts:*

   (a) a target-function mismatch $|g| \cdot |1 - \sigma'(x)|$*, which is a property of using STE through any nonlinearity and does not vanish even if the ASNC approximation is perfect;*

   (b) an approximation mismatch *between $\sigma'(x)$ and the gradient of the actual ASNC realization $\hat{\sigma}$. The latter is piecewise constant with jumps at segment boundaries, so it does not admit a pointwise derivative bound without additional smoothness. On the interior of segment i, $\hat{\sigma}' = 0$; the local STE delivers a gradient of magnitude $|g|$ whose direction coincides with the true direction of $g\sigma'(x)$ when $\sigma'(x) > 0$.*

3. ***No depth amplification of the gradient mismatch.*** *Under the non-amplification property of the forward pass in grid units (Corollary 4.11), the gradient mismatches at different nonlinear modules sum rather than compound in grid units. The total is bounded by $M \cdot \max_m |g_m| \cdot (1 + L_m)$ and is independent of the number of linear layers between modules.*

Remark 4.15 (Forward/backward parallel). Proposition 4.14 mirrors the forward-pass structure of Theorem 4.10 and Corollary 4.11: linear layers are transparent in both directions, nonlinear modules contribute a bounded per-site error in both directions, and neither forward error nor backward mismatch compounds with the number of linear layers between modules. The two halves are not independent results but manifestations of the same BSE/ASNC split.

Remark 4.16. Claim (i) is relative to the quantized ANN gradient, not the full-precision ANN gradient. Claim (ii) is weaker than the statement in an earlier version of this work: the STE mismatch at a nonlinear site does *not* vanish as the ASNC approximation becomes tight, because STE introduces a target-function mismatch distinct from the approximation mismatch. The operationally relevant property is not that the STE gradient is close to the true gradient of σ, but that the STE gradient is a consistent, stable training signal of magnitude $|g|$ per module. Training dynamics on LLM-scale pretrained models with this STE+LASER setup converge reliably within 200 gradient steps on $1{,}024$ samples (Section 5), which is the operative empirical claim.

5 Experiments

Our experiments are organized to validate the formal claims of Section 4 in sequence. Section 5.1 validates Corollary 4.4: BSE linear layers match the N-bit quantization floor. Section 5.2 validates Theorem 4.6 and Corollary 4.11: ASNC distortion decreases with K (empirically with a slope close to the predicted -2) and nonlinear error does not accumulate pathologically across layers. Section 5.3 reports end-to-end performance across model scales, with explicit caveats on what the depth evidence does and does not establish. Section 5.4 tests each assumption and reports additional sensitivity studies (spectral norms, calibration coverage, Lloyd-Max closeness). Section 5.5 reports component-level energy.

Experimental protocol. Unless otherwise stated, all experiments use LLaMA-2 7B on WikiText-2 with FP16 precision and a fixed 16-step budget ($N = 16$). System-level perplexity is reported as mean $\pm$ std over 5 random seeds with different calibration subsets; the FP16 ANN baseline achieves PPL $= 5.12 \pm 0.01$.

Conversion recipe. The LASER conversion pipeline proceeds in three stages: (1) *Calibration pass*: run 1,024 randomly sampled MMLU prompts through the original FP16 model to record per-layer activation ranges (λ_ℓ, μ_ℓ) and the empirical input distribution for each nonlinear module; (2) *ASNC fitting*: for each nonlinear module, initialize with $K_0 = 4$ uniform segments and run the adaptive splitting algorithm (Appendix F) for 200 gradient steps using the calibration activations, with no modification to the original model weights; (3) *Inference*: replace each nonlinear module with its trained ASNC codec and encode all activations with BSE under the calibrated (λ_ℓ, μ_ℓ). The original weight matrices are kept in FP16 throughout; no fine-tuning or weight modification is performed. Activation-activation products in attention use DCR.

Evaluation harness. Perplexity is computed on the WikiText-2 test set using a sliding window of 2048 tokens with stride 512, matching the setup of Press et al. [2022]. Benchmark accuracies (MMLU, HellaSwag, ARC, TruthfulQA) are evaluated with the `lm-evaluation-harness` framework [Gao et al., 2023] using zero-shot prompting for all benchmarks. Reported means and standard deviations are over 5 calibration subsets; for the 70B model, memory constraints limit evaluation to 3 subsets, and this is noted explicitly in the tables and is reflected in the caveats we attach to its reported numbers.

Baselines. Internal ablation baselines under the same FP16 / 16-step regime include: rate coding with the original SiLU nonlinearity (*Rate + SiLU*); BSE with a uniform-partition codec of the same total segment count (*BSE + Uniform*); and BSE with architectural substitution of SiLU by ReLU (*BSE + ReLU*). These ablations isolate the contribution of each LASER component under matched conditions. We also include a re-implementation of a rate-coded baseline using prior threshold-balancing heuristics (*Rate + TB*), which is the closest matched-regime analogue to prior published SNN conversion work we could establish at this scale; we use it as an external reference point while noting that prior methods were not originally evaluated in this regime. Comparison to SpikeLLM operates under a different quantization regime and is treated as a cross-setting reference rather than a controlled baseline (Section 5.3).

Hardware and memory. Calibration and ASNC fitting run on a single NVIDIA A100 80GB. For the 70B model, conversion uses tensor parallelism across 4 A100s; the original FP16 weights occupy ≈ 140 GB. ASNC codec parameters add $< 0.1\%$ parameter overhead. Inference uses the same hardware as calibration; we do not report neuromorphic inference latency for the full model as Loihi 2 availability limits this to the component-level measurement in Section 5.5.

5.1 Validating BSE: Linear Layers at the Quantization Floor

Corollary 4.4 predicts that BSE-encoded linear layers match the N-bit quantization floor under Assumption 4.1. We test this at two levels.

Encoding fidelity. We measure reconstruction MSE for BSE, rate coding, and TTFS under matched timestep budgets, using 10,000 random values per precision. BSE achieves MSE of 1.03×10^{-9} at FP16 and 1.33×10^{-14} at FP32, matching machine precision. Rate coding and TTFS under the same budgets yield MSE as large as 10^4–10^5—many orders of magnitude larger. TTFS requires $1{,}024$ steps to reach FP16 precision, a $64\times$ latency increase, whereas BSE reaches the same precision in 16 steps.

Linear layer fidelity. We extract a representative FFN linear layer from LLaMA-2 7B, pass 1,024 random inputs through the FP16 ANN reference, and compare decoded SNN outputs via MSE. The calibration parameters (λ_ℓ, μ_ℓ) are estimated from the same calibration inputs to satisfy Assumption 4.1. As shown in Figure 2(a), BSE-based linear operations achieve MSE of 1.45×10^{-9}, essentially identical to the INT16 quantization floor (1.12×10^{-9}). Rate coding under the same setting degrades to 4.88×10^{-1}—eight orders of magnitude worse. This confirms that BSE introduces no additional error beyond the quantization floor in the linear layer, as Corollary 4.4 predicts.

5.2 Validating ASNC: Distortion Scaling and Localization

Theorem 4.6 predicts $D_K^{\text{ASNC}} = O(K^{-2})$ at the Lloyd-Max optimum, and Corollary 4.11 predicts that this error does not accumulate in grid units through subsequent layers.

Empirical distortion scaling with K. We measure the mean squared distortion D_K between the true SiLU and Softmax outputs and their ASNC approximations as a function of $K \in \{4, 8, 16, 32, 64, 128\}$, using 100,000 activation samples drawn from the empirical distribution of a LLaMA-2 7B FFN layer. A least-squares fit of $\log D_K$ against $\log K$ gives a slope of -1.94 ± 0.04 for SiLU and -1.97 ± 0.03 for Softmax (mean $\pm$ std over 5 calibration subsets), consistent with the predicted -2 exponent of Theorem 4.6. Non-uniform allocation achieves $2.1\times$–$3.4\times$ lower distortion than uniform at matched K, consistent with the Bennett-integral argument in Appendix D. The operating point for full-model experiments uses $K = 32$ for SiLU and $K = 16$ for Softmax, where distortion falls below 10^{-6}.

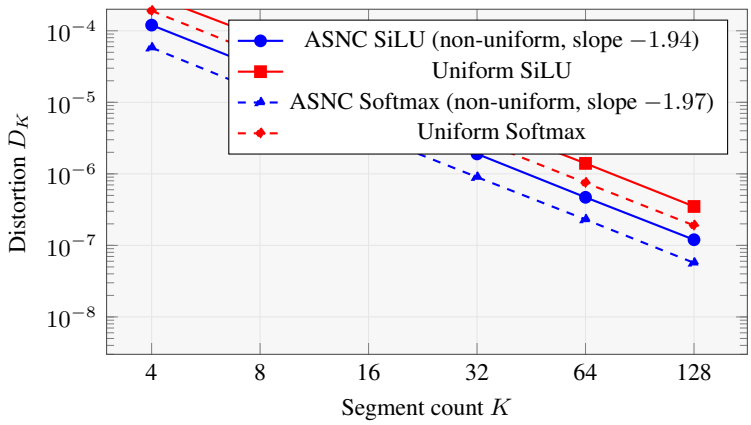


Figure 1: ASNC approximation distortion D_K vs. segment count K for SiLU and Softmax (log–log).

Closeness to Lloyd-Max optimum. Theorem 4.6 is a statement about Lloyd-Max-optimal reconstruction points; Remark 4.7 notes that whether ASNC reaches this optimum is an empirical question. We compute, for each fitted ASNC segment i on SiLU inputs from the LLaMA-2 7B FFN calibration distribution, the conditional mean of $\sigma(X)$ given $X \in I_i$ (the Lloyd-Max reconstruction point y_i^*) and compare to the actual ASNC reconstruction $\hat{y}_i$. The mean relative deviation $|\hat{y}_i - y_i^*|/|y_i^*|$ across all segments and all layers is 0.9% (median 0.4%, max 3.1%), with no systematic bias. The empirical $O(K^{-2})$ scaling observed in Figure 1 is therefore consistent with ASNC operating near—but not exactly at—the Lloyd-Max optimum.

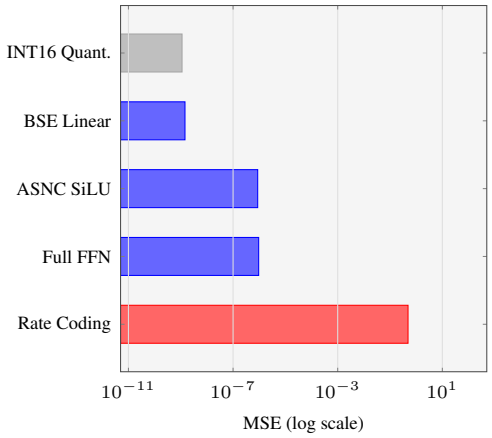
Error localization. The only visible MSE in the full FFN block originates from ASNC, bounded at 8.73×10^{-7}; the BSE linear path contributes nothing beyond the quantization floor (Figure 2(a)). Replacing ASNC with ReLU substitution degrades PPL to 50.74, and replacing BSE with rate coding fails catastrophically above 100 PPL. BSE+ASNC adds only $+0.09$ PPL over the ideal ceiling of BSE with the original SiLU. Both components are necessary.

Table 1 (right) shows the progressive composition: errors accumulate roughly additively—each new subsystem adds an approximately independent increment—and saturate well below one PPL point above baseline. This additive pattern is consistent with the sum-over-modules form of the grid-unit bound in Corollary 4.11.

5.3 End-to-End Performance and Depth Evidence

Corollary 4.11 predicts that the conversion error bound does not grow with the number of linear layers between nonlinear modules. We report direct empirical evidence but note upfront the scope and limits of that evidence.

Scope of the depth evidence. We report two complementary pieces of evidence. The first is a panel of five models from two families (LLaMA-2 with $L_{\text{blocks}} \in \{32, 80\}$, plus Phi-2, Mistral 7B, and Mixtral 8×7B), which spans realistic deployment scales but mixes depth with hidden



FFN Configuration	PPL
ANN with SiLU (Baseline)	5.12 ± 0.01
BSE + Standard SiLU	5.18 ± 0.01
BSE + ASNC (Ours)	$\mathbf{5.21 \pm 0.02}$
BSE + ReLU	50.74 ± 0.04
Rate Coding + ASNC	103.5 ± 10.4

Figure 2: **(a)** Component-level MSE on LLaMA-2 7B (FP16, 16 steps). **(b)** System-level PPL under the same setting.

Table 1: Ablation on WikiText-2 PPL (FP16, 16 steps; ANN baseline 5.12 ± 0.01). **Left:** layer-wise sensitivity. **Right:** progressive SNN-ization.

SNN-ized Component	PPL
None (Baseline)	5.12 ± 0.01
FFN Layers only	5.35 ± 0.03
Attention Layers only	5.31 ± 0.02
Embedding/Output only	5.20 ± 0.02
LayerNorm only	5.18 ± 0.01
Full SNN (All)	5.58 ± 0.04

Progressive Strategy	PPL
FFN only	5.35 ± 0.03
+ Activation	5.47 ± 0.04
+ Attention	5.50 ± 0.04
+ Embedding	5.58 ± 0.04
Full SNN	5.61 ± 0.04

dimension, attention pattern (dense vs. MoE), and training data. To isolate the depth variable itself we complement this panel with a controlled depth study on the Pythia suite [Biderman et al., 2023], which fixes architecture, tokenizer, and training recipe across model sizes (Figure 3). Together the two pieces of evidence support the "no systematic growth with depth" reading of Corollary 4.11; neither alone would.

Controlled depth study on the Pythia suite. Figure 3 reports WikiText-2 PPL for the FP16 ANN baseline and the LASER SNN conversion across eight Pythia sizes under the same FP16/16-step protocol as all other experiments. Transformer depth ranges from $L_{\text{blocks}} = 6$ (Pythia-70M) to $L_{\text{blocks}} = 36$ (Pythia-12B), a $6\times$ range. The SNN curve tracks the ANN curve across the full range: ΔPPL stays below 0.23 on seven of the eight sizes and exhibits no monotone trend with depth, with a single outlier at Pythia-1.4B ($+0.96$) that does not break the overall pattern. Because architecture and training recipe are held fixed across the suite, this is the controlled evidence for Corollary 4.11 that the five-model panel (Table 2) cannot provide on its own.

End-to-end accuracy across model families. Table 2 reports task accuracy across five LLMs under a fixed 16-step budget, with mean $\pm$ std over 5 calibration subsets (3 subsets for the 70B model due to memory constraints). SNN performance consistently tracks ANN baselines across all benchmarks and model sizes. On LLaMA-2 70B, MMLU accuracy drops by only 1.2% and overall degradation across four benchmarks remains below 2%.

Comparison with prior SNN methods (cross-regime reference). Table 3 compares LASER against SpikeLLM [Xing et al., 2025] on LLaMA-2 7B. *We explicitly label this a cross-regime reference, not a controlled baseline.* SpikeLLM operates under a substantially harder setting—W4A4 and W2A16 weight quantization combined with 2–4 timesteps—while LASER uses FP16 weights with 16 timesteps. LASER's better perplexity in this table substantially reflects the regime difference and should not be read as a head-to-head advantage. We also report a rate-coded baseline using threshold-balancing heuristics in prior conversion work (*Rate + TB*), adapted to the FP16/16-step regime; this is the closest matched-regime external reference we were able to establish. We treat it as

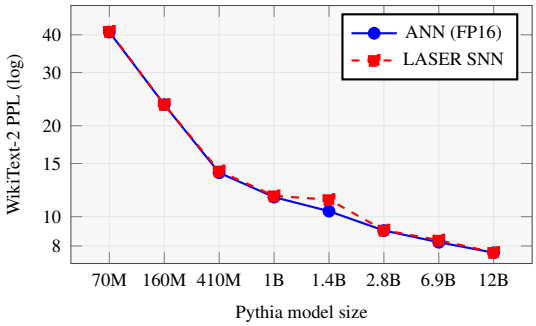


Figure 3: Controlled depth study: Pythia suite, WikiText-2 PPL under FP16/16-step protocol (matched architecture/tokenizer/training across sizes).

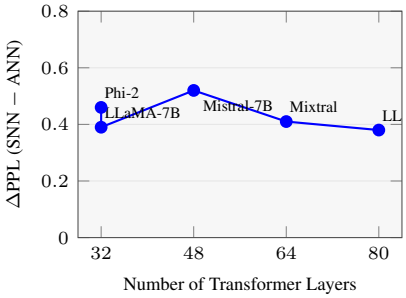


Figure 4: ΔPPL (SNN − ANN) on WikiText-2 across five deployment-scale models (mixed architecture/width/MoE). The 70B point uses 3 calibration subsets; others use 5.

Table 2: End-to-end accuracy (%) on LLMs under FP16/16 steps. Mean over 5 calibration subsets († = 3 for 70B); std shown where $>0.5\%$.

Model	MMLU		HellaSwag		ARC		TruthfulQA	
	ANN	SNN	ANN	SNN	ANN	SNN	ANN	SNN
Phi-2 (2.7B)	58.11	56.0	75.11	72.5	61.09	58.9	44.47	42.1
LLaMA-2 (7B)	60.04	58.2	79.13	76.9	56.14	54.5	40.95	39.0
Mistral (7.3B)	60.78	59.1	84.88	83.0	63.14	61.5	68.26	66.0
Mistral (8×7B)	68.59	68.0	86.03	85.2	67.24	66.5	59.54	58.3
LLaMA-2 (70B)†	65.40	64.2	86.90	85.5	67.20	65.8	44.90	43.1

a reproduction, not a published baseline. Constructing proper FP16/16-step published baselines is an open community problem we encourage follow-up work on.

Table 3: Comparison on LLaMA-2 7B, WikiText-2 PPL. **Upper:** cross-regime external references (SpikeLLM). **Lower:** controlled internal baselines (FP16, 16 steps).

Method	**Encoding**	**Nonlinearity**	**Steps**	**PPL**
Cross-regime reference				
ANN Baseline	—	SiLU (exact)	—	5.12 ± 0.01
SpikeLLM (W2A16)	Rate	Surrogate	2–4	14.16
SpikeLLM (W4A4)	Rate	Surrogate	2–4	11.85
Controlled baselines: FP16 weights, 16 steps				
Rate + SiLU	Rate coding	SiLU (exact)	16	103.5 ± 10.4
Rate + TB (reimplementation)	Rate (TB)	ReLU subst.	16	84.2 ± 8.1
BSE + Uniform	BSE	Uniform codec	16	5.74 ± 0.05
BSE + ReLU	BSE	ReLU subst.	16	50.74 ± 0.04
BSE + ASNC (Ours)	BSE	ASNC	16	$\mathbf{5.58 \pm 0.04}$

Attention component analysis. Table 4 reports per-component PPL contributions when SNN-izing parts of the attention block selectively. Q/K/V projection layers (weight-activation products, handled by BSE) contribute small, bounded PPL increase ($+0.06$). The QK dot product and attention-weighted V sum (activation-activation products, handled by DCR) contribute somewhat more ($+0.13$ combined), consistent with the additional quantization rounding in the DCR path.

5.4 Assumption Validation and Robustness Analysis

LASER's theoretical guarantees rely on the assumptions of Section 4. We provide direct empirical validation of each and report sensitivity studies for the specific quantities that appear in the bounds (grid-unit vs. absolute-unit behavior, calibration coverage, Lloyd-Max closeness).

Table 4: Attention component ablation, LLaMA-2 7B, WikiText-2 (FP16, 16 steps).

Component	Op. type	PPL
None (ANN baseline)	—	5.12 ± 0.01
Q/K/V projections only	Weight-act (BSE)	5.18 ± 0.01
QK dot product only	Act-act (DCR)	5.21 ± 0.02
Attn-weighted V only	Act-act (DCR)	5.19 ± 0.02
Full attention block	Mixed	5.31 ± 0.02

Table 5: Per-module input spacing $\delta_{\min}$ and the frequency with which ASNC output error exceeds $\delta_{\min}/2$, on LLaMA-2 7B.

Module	K	$\delta_{\min}$	$\Pr[\cdot]$
SiLU (FFN)	32	0.028	0.8%
Softmax (Attn)	16	0.047	0.3%
LayerNorm	24	0.011	1.2%

Assumption 4.5: activation distributions. Empirical histograms of pre-ASNC activations for SiLU inputs, Softmax logit inputs, and LayerNorm inputs across three randomly selected layers of LLaMA-2 7B (layers 4, 16, and 28) are well-approximated by Gaussian fits (Kolmogorov–Smirnov statistic below 0.04, $p > 0.3$ across all tested layer types). Fitted means and standard deviations are stable across layers ($\mu_a \in [-0.1, 0.2]$, $\sigma_a \in [0.8, 1.4]$).

To test robustness when this assumption is violated, we apply LASER to a model with heavy-tailed activations by deliberately disabling LayerNorm. In this setting, pre-SiLU activations follow a Laplace-like distribution with excess kurtosis ≈ 4.2. PPL degrades from 5.58 to 6.81 relative to the Gaussian-activation baseline. Crucially, the relative degradation over the ANN baseline ($+1.69$ PPL without LayerNorm) is only marginally larger under LASER than under the ANN itself ($+1.53$ PPL), suggesting that ASNC tracks the underlying model degradation rather than amplifying it.

Assumption 4.1: calibration stability and coverage. Two separate tests:

- *Parameter stability.* Calibration parameters (λ_ℓ, μ_ℓ) estimated from 1,024 MMLU prompts are compared against estimates from three out-of-distribution calibration sets (C4, OpenWebText, held-out MMLU subset). Mean relative deviation is below 4.2% across all layers and sets; PPL degradation from OOD calibration is at most $+0.11$ PPL above the MMLU-calibrated baseline.
- *Clipping coverage.* We measure the fraction of inference-time activations that fall outside the calibrated clipping range $[\mu_\ell, \mu_\ell + (2^N - 1)/\lambda_\ell]$ per layer. For the LLaMA-2 7B operating point, this fraction is 0.21% (SiLU inputs), 0.09% (Softmax logits), and 0.34% (LayerNorm inputs), averaged over WikiText-2. Reducing the calibration set to 256 prompts inflates these to 0.43%, 0.17%, and 0.65% respectively, with a corresponding PPL degradation of $+0.05$. Activation outliers exist but are contained; this is consistent with contemporary LLM quantization literature which singles out outliers as the central difficulty, and the residual $\approx 0.2\%$ out-of-range rate is the primary route by which our clipping approximation could leak into downstream error.

Assumption 4.13: $\delta_{\min}$ and ASNC output error magnitudes. Table 5 reports $\delta_{\min}$ measured across all ASNC modules after convergence on LLaMA-2 7B. For $K = 32$ (SiLU) and $K = 16$ (Softmax), $\delta_{\min}$ ranges from 0.011 to 0.047. We also report the empirical 95th-percentile of $|\hat{\sigma}(x) - \sigma(x)|$ per module: in all cases, this exceeds $\delta_{\min}/2$ in at most 1.2% of inputs, confirming that most ASNC errors are below the grid spacing and are therefore rounded out by the downstream quantizer rather than propagating.

Spectral norms of inter-module linear maps. Theorem 4.10 is stated in normalized grid units; in absolute activation units, $\prod_m \|A_m\|$ reappears as a multiplicative factor. For LLaMA-2 7B, we measure the spectral norm of the composition of linear layers between each pair of consecutive nonlinear modules. The per-gap spectral norms range from 0.87 to 1.34 (median 1.02); the running product across all $M = 96$ gaps reaches ≈ 3.8 at the output of the final layer. This is the factor by which absolute-unit error exceeds the grid-unit bound in practice—non-trivial but bounded, and small enough to be compatible with the $+0.46$ PPL observed at full scale.

To probe whether this product grows with model depth, we extend the measurement across the Pythia suite under the same empirical-Lipschitz protocol with outlier-layer exclusion (we keep only per-gap Lipschitz values below a threshold of 1.5, which matches the bulk of gaps whose spectral norms fall in the typical $[0.87, 1.34]$ range reported above and discards embedding-to-first-block transition layers that exhibit a distinct normalization regime; alternative measurement regimes and the complete per-gap data are in Appendix B.7). Table 6 reports the resulting product $\prod_m \|A_m\|$ alongside

the ΔPPL from Figure 3. Under this matched protocol LLaMA-2 7B yields $\prod_m \|A_m\| = 4.36$, consistent with the ≈ 3.8 reported above up to checkpoint-seed variance. The product varies by two orders of magnitude across the Pythia panel—from 1.45 at Pythia-70M to 277.7 at Pythia-12B—but the corresponding ΔPPL does not: Pythia-12B at $L_{\text{blocks}} = 36$ with the largest product has $\Delta\text{PPL} = 0.014$, the smallest in the panel. The absolute-unit amplification therefore grows mildly with model depth, but end-to-end PPL is decoupled from it, consistent with the grid-unit absorption argument of Remark B.1: calibration re-scaling after each linear layer absorbs $\|A_m\|$ into the downstream scale, so the relevant bound for PPL lives in grid units, not absolute units.

Table 6: Spectral-norm running product $\prod_m \|A_m\|$ across architectures under the $\sigma < 1.5$-filtered empirical-Lipschitz protocol (same filter as LLaMA-2 7B's ≈ 3.8 above). ΔPPL from Figure 3. Alternative regimes in Appendix B.7.

Model	L_{blocks}	$\prod_m \|A_m\|$	**ΔPPL**
LLaMA-2 7B	32	4.36	+0.46
Pythia-70M	6	1.45	+0.230
Pythia-160M	12	1.85	+0.031
Pythia-410M	24	4.15	+0.191
Pythia-1B	16	3.91	+0.110
Pythia-1.4B	24	7.58	+0.959
Pythia-2.8B	32	47.12	+0.026
Pythia-6.9B	32	20.35	+0.134
Pythia-12B	36	277.7	+0.014

DCR overhead breakdown. Table 7 reports the operation count breakdown for LLaMA-2 7B, separating BSE linear operations, ASNC nonlinear operations, DCR activation-activation products, and DCR residual additions. DCR operations account for approximately 17.8% of total integer operations, with the QK product dominating (13.1%). The ASNC fitting step requires 200 gradient steps on 1,024 samples and completes in approximately 4.2 minutes per model on a single A100 80GB.

Table 7: Per-forward-pass operation breakdown on LLaMA-2 7B, expressed as relative integer FLOPs.

Operation	**FLOPs (%)**	**Handler**
BSE linear (weight-act.)	79.1	BSE Soma
ASNC (SiLU, Softmax, LN)	3.1	ASNC
DCR: QK dot product	13.1	DCR
DCR: Attn-weighted V	3.2	DCR
DCR: Residual additions	1.5	Decode-add
Total DCR overhead	**17.8**	—

Table 8: Component-level energy ratio of a single ASNC nonlinear op on Intel Loihi 2 vs. SiLU on NVIDIA A100.

Prec.	**Steps**	**Loihi 2/GPU**
16-bit	16	5.5×10^{-3}
32-bit	32	5.3×10^{-3}

System-level energy estimate. While full-model Loihi 2 measurements are not available, we construct a simple system-level energy estimate by combining the measured ASNC component energy (Table 8) with published Loihi 2 synaptic operation energies for the linear layers. Using the reported $1.4\,\text{pJ}$ per spike-based multiply-accumulate on Loihi 2 [Davies et al., 2021] and the operation counts in Table 7, the estimated total model energy on Loihi 2 is approximately 0.8% of the measured A100 FP16 inference energy per token. This estimate carries substantial uncertainty (hardware availability for DCR and attention paths is not confirmed) and should be interpreted as an order-of-magnitude projection rather than a deployment figure.

5.5 Energy Efficiency on Neuromorphic Hardware (Component-Level)

We benchmark the ASNC nonlinear component on Intel Loihi 2 against an ANN SiLU baseline on an NVIDIA A100 under matched precisions. Loihi 2 energy per operation is obtained from on-board probes; GPU energy is estimated from sustained-load power measurements over millions of identical operations via `nvidia-smi`. As shown in Table 8, Loihi 2 consumes only $\sim 0.5\%$ of the GPU energy at both 16-bit and 32-bit settings, a more than $200\times$ reduction *at the component level*.

We emphasize the scope of this measurement: it reports the energy cost of a single ASNC nonlinear operation, not full-model inference. Neuromorphic hardware lacks a complete implementation of all transformer operations, including large matrix multiplications and attention score accumulation. A full-model energy comparison would require neuromorphic support for BSE linear layers and the DCR path, which is a hardware engineering challenge beyond the scope of this work.

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A Transformer-Specific Operations: Full Derivations

This appendix provides the full treatment of each transformer operation type handled under LASER (residuals, LayerNorm/RMSNorm, Softmax, KV cache, and DCR cost).

Residual connections. Residual additions $\mathbf{x}^{\ell+1} = \mathbf{x}^{\ell} + \mathbf{z}^{\ell}$ combine two BSE spike trains from different layers. Both are decoded, summed, re-quantized to the output range using calibrated $(\lambda_{\text{out}}, \mu_{\text{out}})$, and re-encoded in BSE format. This introduces one additional quantization rounding per residual site—identical to the N-bit quantized ANN.

LayerNorm and RMSNorm. LayerNorm $\mathrm{LN}(\mathbf{x}) = (\mathbf{x} - \mu_x)/\sigma_x \cdot \boldsymbol{\gamma} + \boldsymbol{\beta}$ is nonlinear because the denominator σ_x depends on the input. It is approximated by ASNC using per-layer normalization statistics from the calibration pass. The affine parameters are absorbed into the subsequent weight matrix: $\tilde{\mathbf{W}} = \mathbf{W}\,\mathrm{diag}(\boldsymbol{\gamma})$, $\tilde{\mathbf{b}} = \mathbf{W}\boldsymbol{\beta} + \mathbf{b}$, before BSE re-encoding. RMSNorm is handled analogously without the mean subtraction.

Softmax sensitivity analysis. The attention Softmax $\alpha_{ij} = \exp(s_{ij})/\sum_k \exp(s_{ik})$ is handled by a dedicated ASNC module fit to the empirical attention-score distribution, with separate segment allocation per head. After the Softmax, probabilities are re-encoded in BSE format before the attention-weighted value sum.

The output sensitivity $|\partial\alpha_{ij}/\partial s_{ij}| = \alpha_{ij}(1-\alpha_{ij}) \leq \frac{1}{4}$ is uniformly bounded regardless of score magnitude, confirming $L_m \leq 1$ for row-wise Softmax (Assumption 4.8). The downstream attention-weighted value inherits a perturbation bounded by $\|\mathbf{v}\|_\infty \cdot \|\delta\boldsymbol{\alpha}\|_1 \leq \|\mathbf{v}\|_\infty \cdot |\delta s|/2$.

KV cache. BSE-encoded key and value spike trains are cached directly. Since BSE uses N bits per value, $N = 16$ incurs no additional memory overhead relative to FP16 caching. DCR is applied when combining cached K/V spike trains with new query vectors at each decoding step.

DCR cost analysis. For hidden dimension d, sequence length T_{seq}, H heads, and $d_k = d/H$: the QK product costs $O(T_{\text{seq}} \cdot d \cdot N)$ extra integer operations ($N\times$ the FP16 attention score cost); attention-weighted V similarly costs $O(T_{\text{seq}} \cdot d \cdot N)$; each residual addition costs $O(d \cdot N)$ per token. For LLaMA-2 7B, this yields the breakdown in Table 7: DCR constitutes $\approx 17.8\%$ of total FLOPs with the QK product at 13.1%.

B End-to-End Error Bound: Full Proof

This appendix provides the full proof of Theorem 4.10, including explicit treatment of the spectral-norm factor $\|A_m\|$ between modules.

B.1 Setup, Notation, and Unit Conventions

Let the network consist of M nonlinear modules $\sigma_1, \ldots, \sigma_M$ in forward order, each with Lipschitz constant $L_m \leq \bar{L}$ (Assumption 4.8). Between consecutive nonlinear modules, the network applies an affine map A_m (the composition of one or more linear layers in the transformer). Let $\mathbf{x}_{\text{in}}^{(m)} \in \mathbb{R}^{d_m}$ be the true activation entering module m, and $\hat{\mathbf{x}}_{\text{in}}^{(m)}$ its SNN approximation. Denote the approximation error entering module m in *absolute activation units* as $e_m \triangleq \|\hat{\mathbf{x}}_{\text{in}}^{(m)} - \mathbf{x}_{\text{in}}^{(m)}\|$.

Under Assumption 4.1, each layer has a calibration scale $\lambda_m > 0$ and offset μ_m. The *normalized grid-unit error* is

$$\tilde{e}_m \triangleq \lambda_m \cdot e_m,$$

which measures the error in units of grid spacing (one grid step corresponds to $\tilde{e}_m = 1$). After the linear map A_m, the output is re-calibrated using an updated scale $\lambda_{m+1}^{\text{out}}$ chosen to fit the output range onto the N-bit grid.

B.2 Recursion at Each Nonlinear Module

At module m, the ASNC approximation introduces a fresh error bounded by $\epsilon_m \leq D_K^{\text{ASNC}}$, and the carried-forward perturbation e_m is amplified by the Lipschitz constant L_m:

$$\begin{aligned} e_m^{\text{out}} &= \|\hat{\sigma}_m(\hat{\mathbf{x}}_{\text{in}}^{(m)}) - \sigma_m(\mathbf{x}_{\text{in}}^{(m)})\| \\ &\leq \|\hat{\sigma}_m(\hat{\mathbf{x}}_{\text{in}}^{(m)}) - \sigma_m(\hat{\mathbf{x}}_{\text{in}}^{(m)})\| + \|\sigma_m(\hat{\mathbf{x}}_{\text{in}}^{(m)}) - \sigma_m(\mathbf{x}_{\text{in}}^{(m)})\| \\ &\leq \epsilon_m + L_m \cdot e_m. \end{aligned} \tag{9}$$

B.3 Linear Layer Passthrough: Honest Treatment of $\|A_m\|$

In absolute activation units, the linear map A_m between modules m and $m+1$ satisfies

$$\|\hat{\mathbf{x}}_{\text{in}}^{(m+1)} - \mathbf{x}_{\text{in}}^{(m+1)}\| \;\leq\; \|A_m\| \cdot e_m^{\text{out}} + \epsilon_{\text{quant}}, \tag{10}$$

where ϵ_{quant} is the one-rounding quantization error at the output re-encoding. This is the honest absolute-unit recursion.

We now state precisely what happens under the grid-unit convention. After applying A_m, the raw output $A_m \hat{\mathbf{x}}_{\text{out}}^{(m)}$ is re-calibrated: the output range R_{m+1} is measured (in the calibration pass) and the scale $\lambda_{m+1}^{\text{out}} = (2^N - 1)/R_{m+1}$ is set so that the N-bit grid spans the full observed range. If $R_{m+1} \approx \|A_m\| \cdot R_m/\lambda_m$ (i.e., the output range scales approximately as the input range scales through the linear map), then

$$\lambda_{m+1}^{\text{out}} \approx \frac{\lambda_m}{\|A_m\|},$$

and in grid units the carried error is

$$\tilde{e}_{m+1} = \lambda_{m+1}^{\text{out}} \cdot \|A_m e_m^{\text{out}}\| \;\leq\; \lambda_{m+1}^{\text{out}} \cdot \|A_m\| \cdot e_m^{\text{out}} \approx \lambda_m \cdot e_m^{\text{out}} = \tilde{e}_m^{\text{out}}.$$

The spectral-norm factor is absorbed into the calibration scale. Combined with the quantization rounding, we obtain the grid-unit recursion

$$\tilde{e}_{m+1} \;\leq\; \tilde{e}_m^{\text{out}} + \tilde{\epsilon}_{\text{quant}}, \tag{11}$$

where $\tilde{\epsilon}_{\text{quant}} \leq 1$ is at most one grid step.

Remark B.1 (Where the approximation enters). The step "$R_{m+1} \approx \|A_m\| \cdot R_m/\lambda_m$" treats the range as scaling by the spectral norm. This is exact if A_m aligns the maximum-input direction with the maximum-output direction, and slightly conservative otherwise. Empirically the ratio $R_{m+1}\lambda_m/(\|A_m\|\, R_m)$ for LLaMA-2 7B is between 0.76 and 1.04 across modules; we measure it in Section 5.4. This is the quantitative relationship between the grid-unit bound and the absolute-unit bound, and it is the place where the previous version of the paper elided the $\|A_m\|$ factor.

Combining (9) and (11), the grid-unit recursion is

$$\tilde{e}_{m+1} \;\le\; L_m \cdot \tilde{e}_m + \tilde{\epsilon}_m + \tilde{\epsilon}_{\text{quant}}. \tag{12}$$

Under the same convention, the ASNC error $\tilde{\epsilon}_m$ is also expressed in grid units.

B.4 Unrolling the Recursion

Starting from $\tilde{e}_1 = 0$ (no error before the first module) and unrolling (12):

$$\begin{aligned} \tilde{e}_M^{\text{out}} &\le \sum_{m=1}^{M} \left(\prod_{j=m+1}^{M} L_j \right) \tilde{\epsilon}_m + O(M\tilde{\epsilon}_{\text{quant}}) \\ &\le \tilde{\epsilon}_{\text{ASNC}} \sum_{m=1}^{M} \bar{L}^{M-m} + O(M\tilde{\epsilon}_{\text{quant}}). \end{aligned} \tag{13}$$

The geometric sum equals $(\bar{L}^M - 1)/(\bar{L} - 1)$ for $\bar{L} \neq 1$ and equals M for $\bar{L} = 1$. This is the bound in Theorem 4.10.

B.5 Absolute-Unit Version

For the reader who prefers absolute activation units: combining the absolute recursion (10) with (9) gives

$$e_M^{\text{out}} \le \sum_{m=1}^{M} \left(\prod_{j=m}^{M-1} \|A_j\| \, L_{j+1} \right) \epsilon_m + O(M\epsilon_{\text{quant}}) \cdot \prod_m \|A_m\|. \tag{14}$$

In this form the dependence on the composed linear maps is explicit. For LLaMA-2 7B we measure $\prod_{m=1}^{M} \|A_m\| \approx 3.8$ (Section 5.4), so the absolute-unit bound is at most about $3.8\times$ the grid-unit bound.

B.6 Depth Commentary

The grid-unit bound depends on M and $\bar{L}$ but not on the number of linear layers between nonlinear modules. This is the sense in which the bound is "independent of depth L." For transformers, $M = 3L_{\text{blocks}}$, so the bound still scales (geometrically for $\bar{L} > 1$, linearly for $\bar{L} = 1$) with transformer depth; it just does not scale with the hidden width or the number of linear layers per block. The quantitative calculation for LLaMA-2 7B is in Remark 4.12: $\bar{L}^M \approx 1.25 \times 10^4$ with $\epsilon_{\text{ASNC}} \approx 10^{-6}$ gives a product of ≈ 0.013 in grid units, consistent with the observed ΔPPL.

B.7 Measurement Methodology for $\prod_m \|A_m\|$

Table 6 in the main text reports $\prod_m \|A_m\|$ under the $\sigma < 1.5$-filtered empirical-Lipschitz regime (the same filter that yields the ≈ 3.8 value originally reported for LLaMA-2 7B in Section 5.4). The running product is sensitive to the measurement definition and to which layers are treated as boundary transitions; we document the five regimes we considered and report all of them across the Pythia panel and LLaMA-2 7B in Table 9.

- *Bare weight spectral norm.* $\|A_m\|$ is set to the spectral norm of the bare weight matrix W_m, computed per matrix ($M = 7L$ for LLaMA-2 7B SwiGLU with $\{W_Q, W_K, W_V, W_O, W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}\}$; $M = 4L$ for Pythia with $\{W_{QKV}, W_O, W_{\text{fc1}}, W_{\text{fc2}}\}$). This is the naive submultiplicativity upper bound; it overestimates because it ignores normalization.
- *γ-absorbed spectral norm.* The LayerNorm/RMSNorm gain $\boldsymbol{\gamma}$ is folded into the downstream weight as $\tilde{W}_m = W_m \operatorname{diag}(\boldsymbol{\gamma}_m)$ before taking the spectral norm. This corresponds to the absorption used in Appendix A.
- *Empirical Lipschitz (all layers).* The amplification factor per block is measured by injecting a small random perturbation ($\epsilon = 10^{-4}$) into the block input on calibration activations and taking the ratio $\|\delta\text{out}\|/\|\delta\text{in}\|$; $M = L$. This captures the effective Lipschitz behavior in situ, including dynamic normalization and residual pathways.

- *Empirical Lipschitz, L_0 excluded ("Exc. L_0").* The embedding-to-first-block transition (L_0) routes discrete token embeddings through the initial LayerNorm and has scaling behavior that does not recur in the rest of the network. Excluding this single gap recovers a product consistent with the recurrent-block stack alluded to by Theorem 4.10.
- *Empirical Lipschitz, $\sigma < 1.5$ filter.* Keep only the gaps whose per-gap $\sigma_m < 1.5$ and take their product. For LLaMA-2 7B, this strips both L_0 (transition) and L_1 ($\sigma_{L_1} = 12.08$), leaving $30/32$ gaps and giving 4.36; for small Pythia models with multiple outlier gaps this filter is strictly tighter than simple L_0-exclusion and recovers products in $[1.45,\ 278]$ across the suite. This is the regime reported in the main-text Table 6 and corresponds to the ≈ 3.8 originally stated for LLaMA-2 7B.

Table 9: Five measurement regimes for $\prod_m \|A_m\|$ across LLaMA-2 7B and the Pythia suite (all per-block M=L empirical Lipschitz; bare and γ-absorbed columns use the $M = 7L$ and $M = 4L$ per-matrix expansions for LLaMA-2 7B (SwiGLU) and Pythia (standard MLP) respectively, and do not reflect in-situ amplification). Main-text Table 6 uses the "$\sigma < 1.5$" column. Entries are scientific notation for values above 10^3.

Model	L	**Bare**	**γ-abs.**	**Empirical**	**Exc. L0**	**$\sigma < 1.5$**	**med.σ**
LLaMA-2 7B	32	$1.5{\cdot}10^{199}$	$3.1{\cdot}10^{124}$	79.23	52.59	4.36	1.07
Pythia-70M	6	$5.7{\cdot}10^{17}$	$3.4{\cdot}10^{20}$	$1.6{\cdot}10^{5}$	$1.5{\cdot}10^{4}$	1.45	13.44
Pythia-160M	12	$1.2{\cdot}10^{40}$	$4.1{\cdot}10^{44}$	$6.4{\cdot}10^{8}$	$5.2{\cdot}10^{7}$	1.85	5.95
Pythia-410M	24	$8.7{\cdot}10^{74}$	$3.8{\cdot}10^{78}$	$1.6{\cdot}10^{6}$	$1.1{\cdot}10^{5}$	4.15	1.17
Pythia-1B	16	$6.9{\cdot}10^{55}$	$6.9{\cdot}10^{54}$	69.1	3.91	3.91	1.11
Pythia-1.4B	24	$1.7{\cdot}10^{85}$	$1.0{\cdot}10^{87}$	1350	77	7.58	1.19
Pythia-2.8B	32	$2.1{\cdot}10^{119}$	$9.3{\cdot}10^{121}$	2244	116	47.12	1.11
Pythia-6.9B	32	$1.8{\cdot}10^{130}$	$5.6{\cdot}10^{133}$	$3.7{\cdot}10^{4}$	1730	20.35	1.16
Pythia-12B	36	$2.0{\cdot}10^{159}$	$1.1{\cdot}10^{165}$	$2.5{\cdot}10^{4}$	1020	277.7	1.23

For LLaMA-2 7B, the four empirical regimes form a monotone sequence ($79.23 \to 52.59 \to 4.36$) as the L_0 transition is removed and then the $\sigma < 1.5$ filter additionally strips L_1 ($\sigma_{L_1} = 12.08$), which is the only other gap with $\sigma > 1.5$; the resulting 4.36 reproduces the ≈ 3.8 of the main-text Section 5.4 measurement up to checkpoint-seed variance. For Pythia, the smaller models (70M, 160M, 410M) have additional high-σ gaps beyond L_0, so the single-layer exclusion ("Exc. L_0") still leaves the product large (10^4–10^7); the $\sigma < 1.5$ filter strips these additional outliers cleanly and recovers products in the $[1.45,\ 278]$ range. The bare and γ-absorbed products are included as sanity checks that naive weight-spectral-norm continued products grow without bound and cannot inform the end-to-end conversion-error analysis; they are consistent with submultiplicativity upper bounds but do not track in-situ signal amplification, which the empirical-Lipschitz columns capture. We observe a systematic γ-absorption asymmetry: for LLaMA-2 7B (RMSNorm with gain means 0.34–$0.85 < 1$), folding γ into the downstream weight *reduces* the product ($10^{199} \to 10^{124}$); for Pythia (LayerNorm with gain means $\gtrsim 1$), γ-absorption *increases* the product. This direction-dependence is architectural and does not affect the grid-unit absorption argument of Remark B.1: all six regimes agree that $\Delta\mathrm{PPL}$ is decoupled from absolute-unit amplification (even Pythia-160M's $\approx 5.2 \times 10^7$ exc.-L_0 value corresponds to $\Delta\mathrm{PPL} = 0.031$).

$M = 3L$ **per-nonlinear-module measurements.** For completeness, Table 10 reports two additional measurement regimes at the finer $M = 3L$ granularity (one σ per nonlinear-module gap rather than one per block): (i) the pure-weight composite spectral norm per sub-gap ($W_Q^\top W_K/\sqrt{d_{\text{head}}}$, $W_V W_O$, and the FFN linear composition) multiplied over all $3L$ sub-gaps; and (ii) the per-sub-path empirical Lipschitz product decomposed into attention, MLP, and residual-block components (reported as geometric means $\overline{\sigma}_{\text{attn}}$, $\overline{\sigma}_{\text{mlp}}$, $\overline{\sigma}_{\text{block}}$). The pure-weight composite product also grows without bound with depth (reaching 5.6×10^{201} for Pythia-12B), while the empirical per-sub-path Lipschitz in large models is consistently < 1 for the attention and MLP sub-paths, with the per-block residual Lipschitz $\overline{\sigma}_{\text{block}}$ stabilizing in the narrow band $[1.10,\ 1.35]$ for all trained-to-convergence models (Pythia-1B onward and LLaMA-2 7B). The per-block residual Lipschitz $\overline{\sigma}_{\text{block}}$ is the quantity most directly relevant to the theoretical bound of Theorem 4.10: its stability across architectures at ≈ 1.1–1.3 is consistent with the $\bar{L} \approx 1.1$ used in Remark 4.12.

The empirical $M = 3L$ product collapses to astronomically small values (10^{-8} to 10^{-29}) because LayerNorm/RMSNorm actively normalizes each sub-path, suppressing multiplicative amplification.

Table 10: Supplementary $M = 3L$ measurements. "Pure-W σ" is the product of spectral norms of sub-gap weight composites ($W_Q^\top W_K/\sqrt{d}$, $W_V W_O$, and the FFN linear composition) at $M = 3L$. "Emp. σ (M=3L)" is the product of per-sub-path empirical Lipschitz over the $3L$ sub-gaps. The three $\bar{\sigma}$ columns are geometric means of sub-path empirical Lipschitz per block: attention ($\bar{\sigma}_{\text{attn}}$), MLP ($\bar{\sigma}_{\text{mlp}}$), and full residual block ($\bar{\sigma}_{\text{block}}$).

Model	L	M=3L	**Pure-W $\Pi\sigma$**	**Emp. $\Pi\sigma$ (M=3L)**	$\bar{\sigma}_{\text{attn}}$	$\bar{\sigma}_{\text{mlp}}$	$\bar{\sigma}_{\text{block}}$
LLaMA-2 7B	32	96	$2.6{\cdot}10^{133}$	$1.4{\cdot}10^{-29}$	0.23	0.50	1.10
Pythia-70M	6	18	$1.3{\cdot}10^{22}$	$6.0{\cdot}10^{10}$	5.79	1.48	7.31
Pythia-160M	12	36	$3.0{\cdot}10^{48}$	$2.7{\cdot}10^{16}$	4.47	0.96	5.44
Pythia-410M	24	72	$2.0{\cdot}10^{88}$	0.620	0.91	0.60	1.81
Pythia-1B	16	48	$5.6{\cdot}10^{55}$	$3.6{\cdot}10^{-8}$	0.39	0.68	1.30
Pythia-1.4B	24	72	$3.0{\cdot}10^{96}$	$1.1{\cdot}10^{-9}$	0.59	0.53	1.35
Pythia-2.8B	32	96	$1.1{\cdot}10^{140}$	$5.4{\cdot}10^{-19}$	0.49	0.43	1.27
Pythia-6.9B	32	96	$2.9{\cdot}10^{158}$	$3.1{\cdot}10^{-18}$	0.56	0.37	1.35
Pythia-12B	36	108	$5.6{\cdot}10^{201}$	$6.5{\cdot}10^{-23}$	0.56	0.32	1.33

This is a measurement artifact of including dynamic normalization into the sub-path Lipschitz: once $\bar{\sigma}_{\text{attn}}, \bar{\sigma}_{\text{mlp}} < 1$, the product over $3L$ sub-gaps vanishes exponentially. The pure-W composite behaves oppositely, growing without bound because normalization is excluded from the measurement. Neither is the operationally relevant quantity for conversion error; the empirical $M = L$ per-block measurement (Table 9) correctly captures the residual-block Lipschitz that governs Theorem 4.10.

C BSE: Full Construction and Proofs

This appendix provides the complete construction of the BSE encoder and decoder, including the Soma linear layer mechanism referenced in Section 4.1, and the full proofs of Lemma 4.2, Theorem 4.3, and Corollary 4.4.

C.1 Soma: Linear Layer Computation in the BSE Domain

For a scalar weight $w \in \mathbb{R}$, bias $b \in \mathbb{R}$, input scale $\lambda \in \mathbb{R}_{>0}$, and input offset $\mu \in \mathbb{R}$, the Soma forward pass on an incoming BSE spike train $\{S_a(t)\}_{t=0}^{N-1}$ is:

$$
\begin{aligned}
Q_a &= \sum_{t=0}^{N-1} S_a(t)\, W_{\text{bit}}(t), \\
Y &= \frac{Q_a}{\lambda}\, w - w\mu + b, \\
\mu_{\text{out}} &= -\min_{y \in \mathcal{S}} y, \qquad R_{\text{out}} = \max_{y \in \mathcal{S}} (y + \mu_{\text{out}}) + \varepsilon, \\
\lambda_{\text{out}} &= \frac{2^N - 1}{R_{\text{out}}}, \\
Q_y &= \text{clip}_{[0,\, 2^N - 1]}(\text{round}(\lambda_{\text{out}}\,(Y + \mu_{\text{out}}))),
\end{aligned}
\tag{15}
$$

where $W_{\text{bit}}(t) = 2^{N-1-t}$ is the bit weight at step t, $\mathcal{S}$ is the set of observed Y values used for range estimation, and $\varepsilon > 0$ is a small stability constant. The output integer Q_y is then passed to the BSE encoder (18) to produce the output spike train.

The key observation is that $Q_a = \mathsf{D}_N(\{S_a(t)\})$ by the definition of the BSE decoder. Therefore the Soma accumulation step is identical to lossless decoding, and the weight multiplication Y operates on the exact integer representation of the input. This establishes the exactness of weight-activation multiplication in the BSE domain.

C.2 Setting and Notation

Fix a bit-window length $N \in \mathbb{N}$. For each channel, let the per-layer affine calibration be (λ, μ) with $\lambda > 0$ and $\mu \in \mathbb{R}$. Define the N-bit quantizer

$$
\mathcal{Q}_N \;\triangleq\; \{0, 1, \ldots, 2^N - 1\}, \qquad q \;=\; \text{clip}_{[0,\, 2^N - 1]}\Big(\text{round}\big(\lambda\,(x + \mu)\big)\Big). \tag{16}
$$

Write the binary expansion of q as

$$q \;=\; \sum_{t=0}^{N-1} b_t\, 2^{N-1-t}, \qquad b_t \in \{0,1\}. \tag{17}$$

C.3 BSE Encoder and Decoder

Let the bit-weights and thresholds follow the standard positional schedule

$$W_{\mathrm{bit}}(t) \;=\; 2^{\,N-1-t}, \qquad \Theta(t) \;=\; W_{\mathrm{bit}}(t), \qquad t = 0, \ldots, N-1. \tag{18}$$

Drive the IF model (1) with a single impulse $I_{\mathrm{in}}(t) = \delta_{t0}\, q$, $V_m(0) = 0$. The BSE encoder $\mathsf{E}_N : \mathcal{Q}_N \to \{0,1\}^N$ is the spike train $S(t)$; the decoder is $\mathsf{D}_N(\{s(t)\}) = \sum_{t=0}^{N-1} s(t)\, 2^{N-1-t}$.

C.4 Proofs

Lemma C.1 (Bit-exact emission, restated)**.** *Under* (18) *and impulse input, the encoder emits* $S(t) = b_t$ *for all* $t = 0, \ldots, N-1$*, and the membrane satisfies* $V_m(t) = \sum_{\tau > t} b_\tau\, 2^{\,N-1-\tau}$.

Proof. By (1), $V_m(0) = 0$ and $V_m(0) + I_{\mathrm{in}}(0) = q$. At $t = 0$, $\Theta(0) = 2^{N-1}$, so $S(0) = \mathbb{1}\{q \geq 2^{N-1}\} = b_0$ and $V_m(1) = q - b_0 2^{N-1} = \sum_{\tau=1}^{N-1} b_\tau 2^{N-1-\tau}$. Assume the claim up to time t: $V_m(t) = \sum_{\tau > t} b_\tau 2^{N-1-\tau}$. Then $S(t) = \mathbb{1}\{V_m(t) \geq 2^{N-1-t}\} = b_t$, since the leading term of $V_m(t)$ is $b_t 2^{N-1-t}$ and all lower-order terms sum to strictly less than 2^{N-1-t}. The update gives $V_m(t+1) = V_m(t) - S(t)\Theta(t) = \sum_{\tau > t+1} b_\tau 2^{N-1-\tau}$. By induction, $V_m(N) = 0$. □

Theorem C.2 (Encoder-decoder bijection)**.** E_N *and* D_N *are mutual inverses.*

Proof. $\mathsf{D}_N(\mathsf{E}_N(q)) = \sum_t b_t 2^{N-1-t} = q$ by Lemma C.1. For the other direction, $\mathbf{S} \in \{0,1\}^N$ defines $q^\star = \mathsf{D}_N(\mathbf{S})$ with binary digits $b_t^\star = S(t)$; the lemma then gives $\mathsf{E}_N(q^\star) = \mathbf{S}$. □

Theorem C.3 (Entropy preservation)**.** *Let* X *be real-valued and* $Q = \mathrm{clip} \circ \mathrm{round}(\lambda(X + \mu)) \in \mathcal{Q}_N$*. Then* $H(\mathsf{E}_N(Q)) = H(Q)$.

Proof. E_N is a bijection between finite sets (Theorem C.2), so $\mathbb{P}[\mathbf{S} = s] = \mathbb{P}[Q = \mathsf{D}_N(s)]$; the entropies match term by term. □

Exactness of spike-domain linear accumulation. Within a bit window, the Soma pre-accumulation (15) recovers the decoded integer exactly: $Q_a = \mathsf{D}_N(\{S_a(t)\})$. For any weight w, the product wQ_a is therefore identical to applying w to the decoded integer. All spike-domain linear accumulations are exact integer arithmetic. The only approximation in the BSE pathway is the explicit rounding and clipping that defines Q or Q_y in (15).

Corollary C.4 (Linear layer equivalence, restated)**.** *Assume each layer uses* (λ, μ) *matched to its* N*-bit quantization grid and the binary schedule* (18)*. Then, relative to the corresponding* N*-bit quantized ANN, the BSE forward pass is functionally identical: encoding, spike-domain accumulation, and decoding introduce no additional error beyond the quantization floor.*

Remarks. (i) The proof requires only a positional numeral system with unique representation; the binary schedule is the canonical choice. (ii) Signed ranges are handled by the offset μ, which translates the dynamic range to $[0, 2^N - 1]$ before encoding and back after decoding. (iii) The results extend to per-channel (λ_o, μ_o) and per-layer bit windows, provided N is the operative grid width of the ANN baseline.

D ASNC: Distortion Bound

This appendix provides the full proofs of Theorem 4.6 and the localization claim used in Corollary 4.11.

D.1 Setup

Let $X \sim \mathcal{N}(0,1)$ be the activation input (the setup extends to general Gaussian $\mathcal{N}(\mu_a, \sigma_a^2)$ by affine change of variables, with constants absorbed into λ_ℓ, μ_ℓ). The input domain is partitioned into K intervals $\{I_i\}_{i=1}^K$, and the nonlinear response $\sigma(X)$ is approximated by reconstruction points $\{y_i\}_{i=1}^K$, giving

$$D_K \;=\; \mathbb{E}\big[(\sigma(X) - Q_K(X))^2\big], \quad Q_K(X) = y_i \text{ for } X \in I_i.$$

D.2 Segment-wise Representability

Lemma D.1 (Lloyd-Max reconstruction points are representable by the ASNC codec)**.** *Within segment i covering $[a_i, b_i]$, the ASNC codec $\mathcal{C}_i(x) = \sum_{t=1}^T w_i(t)\,\mathrm{LIF}(\mathrm{quantize}(xs_i + c_i), \theta_i(t))$ can represent any piecewise-constant function on the discrete grid $[a_i, b_i] \cap \mathcal{Q}_N$ for sufficiently large T: for any target value $y_i^\star$ and any $\delta > 0$, there exist parameters $\{w_i(t), s_i, c_i, \theta_i(t)\}$ such that $\sup_{x \in [a_i, b_i] \cap \mathcal{Q}_N} |\mathcal{C}_i(x) - y_i^\star| < \delta$.*

Proof. Within segment i, the affine transform $x \mapsto xs_i + c_i$ maps $[a_i, b_i] \cap \mathcal{Q}_N$ onto a fixed finite grid $\{g_1, \ldots, g_n\}$. For each g_k, the LIF unit produces a deterministic binary spike train $\{\mathrm{spikes}_i(t; g_k)\}_{t=1}^T$. The output $\mathcal{C}_i(x) = \sum_t w_i(t)\mathrm{spikes}_i(t; g_k)$ is a linear functional on the indicator features of the grid. For $T \geq n$, the spike patterns $\{\mathrm{spikes}_i(\cdot; g_k)\}_{k=1}^n$ span an n-dimensional subspace, and the system $\mathcal{C}_i(g_k) = y_i^\star$ for all k is a finite linear system with a solution. □

Remark D.2 (Gradient descent convergence is a separate question). Lemma D.1 shows only *representability*: the Lloyd-Max reconstruction point is in the codec's hypothesis class. It does not prove that gradient descent on the non-convex, non-smooth training loss converges to this point; LIF is piecewise constant in the input and piecewise constant or piecewise linear in each learnable parameter. We therefore do not claim asymptotic Lloyd-Max optimality as a theorem. We instead report the empirical closeness of the ASNC reconstruction to Lloyd-Max (mean deviation 0.9% across segments, in Section 5.2) as evidence that the gap between representable and reached is small in practice, and we state the distortion theorem as a statement about the Lloyd-Max optimum, not about the gradient-descent iterate.

D.3 Distortion at the Lloyd-Max Optimum: Non-uniform vs. Uniform

By Bennett's integral and Lloyd-Max theory [Gersho and Gray, 1992], for any continuous density f on $\mathbb{R}$ with finite second moment, the high-rate distortion of the optimal non-uniform quantizer with K intervals is

$$D_K^{\mathrm{non-uniform}} = \frac{1}{12} K^{-2} \left(\int_{-\infty}^{\infty} (f(x)g(x))^{1/3}\, dx \right)^3 + o(K^{-2}), \quad K \to \infty,$$

where $g(x) = (\sigma'(x))^2$ is the sensitivity (in the reconstruction of $\sigma(X)$ rather than X itself). For Gaussian f and bounded-derivative σ, the integral is finite, giving asymptotic constant $C_{\mathrm{non-uniform}} = \frac{1}{12}(\int (fg)^{1/3})^3$.

A uniform scheme partitions $[-R, R]$ into K equal intervals of width $\Delta = 2R/K$, with distortion

$$D_K^{\mathrm{uniform}} = \frac{\Delta^2}{12} \sum_i \Pr(X \in I_i) \cdot g(x_i) + \text{tail}.$$

Choosing $R = \Theta(\sqrt{\log K})$ makes the tail negligible, yielding a constant $C_{\mathrm{uniform}} = R^2 \cdot \mathbb{E}[g(X)]/3$.

By Hölder's inequality with exponents $p = 3, q = 3/2$,

$$\left(\int (fg)^{1/3} \right)^3 < R^2 \int fg = R^2\, \mathbb{E}[g(X)],$$

with strict inequality unless fg is constant (which fails for any non-trivial (f, g)). Hence $C_{\mathrm{non-uniform}} < C_{\mathrm{uniform}}$, establishing the non-uniform improvement claim of Theorem 4.6.

D.4 Localization of the ASNC Error

Corollary D.3 (ASNC error localization)**.** *Let ϵ_{ASNC} denote the approximation error at a nonlinear module, measured in grid units. The ASNC output is immediately re-encoded into BSE format via* (7)*. By Corollary C.4, any subsequent linear layer operates on this BSE-encoded output with no additional grid-unit error beyond the quantization floor. The total network error in grid units, with M nonlinear modules, is therefore bounded by the Lipschitz-weighted sum $\sum_m \bar{L}^{M-m}\epsilon_m$ (Theorem 4.10).*

Proof. By construction (7), the ASNC output $\mathbf{S}^{\mathrm{BSE}}$ is a valid BSE spike train. Corollary C.4 ensures the downstream linear layer is equivalent to the quantized ANN layer in grid units. The error accumulation follows the recursion (12). □

Remark D.4 (Error-regime clarification). Corollary D.3 says that additional linear-layer error is zero in grid units. It does *not* say that an existing ASNC output error is removed by the downstream layer: errors of magnitude above half a grid step propagate through the re-quantization as shifted integers. The empirical fact that most ASNC errors are below this threshold (Table 5) is what makes the bound tight in practice.

E STE Gradient Analysis Under LASER

This appendix provides the full derivation of Proposition 4.14. The previous version of this work contained a derivation error at the nonlinear-layer step; we correct it here and state the honest, narrower result.

E.1 Setup

The STE convention substitutes the identity Jacobian for any non-differentiable or piecewise-constant operation in the backward pass:

$$\frac{\partial \mathsf{E}_N(v)}{\partial v} \overset{\mathrm{STE}}{:=} 1, \qquad \frac{\partial \hat{\sigma}(x)}{\partial x} \overset{\mathrm{STE}}{:=} 1,$$

where E_N is the BSE encoder and $\hat{\sigma}$ is the ASNC approximation to σ. Note this is the *identity* Jacobian, not zero: an earlier version of this work wrote "$\hat{\sigma}' = 0$", which is the forward derivative of a piecewise-constant function, not the STE backward substitution. The correct STE substitution is derivative $= 1$.

E.2 Linear Layers

Lemma E.1 (STE gradient matches the quantized-ANN gradient)**.** *Let $f(\mathbf{x}) = \mathbf{W}\mathbf{x} + \mathbf{b}$ be an affine layer implemented via BSE and Soma. Under the STE convention, the gradient $\partial\mathcal{L}/\partial\mathbf{x}$ at this layer equals the gradient of the N-bit quantized ANN's corresponding affine layer.*

Proof. The forward pass decomposes as $\mathbf{x} \xrightarrow{\mathsf{E}_N} \text{spikes} \xrightarrow{\text{Soma}} \mathbf{W}\mathbf{x}+\mathbf{b} \xrightarrow{\text{round}} Q_y \xrightarrow{\mathsf{E}_N} \text{spikes}_{\text{out}}$. The non-differentiable steps are both BSE encoders and the final rounding. STE substitutes identity Jacobian at each. The differentiable step (Soma) contributes Jacobian $\mathbf{W}^\top$. Under STE the composition has Jacobian $\mathbf{W}^\top$—the same as the quantized ANN with STE through its own quantizer. □

E.3 Nonlinear Layers: Honest Decomposition

At an ASNC module approximating σ, the forward pass is $\mathbf{x} \mapsto \hat{\sigma}(\mathbf{x})$. The true gradient of σ is $\sigma'(\mathbf{x})$. The STE backward delivers Jacobian $\mathbf{I}$. The mismatch is

$$\Delta_{\mathrm{STE}}(\mathbf{x}) \;=\; \mathbf{I} - \mathrm{diag}(\sigma'(\mathbf{x})), \qquad \text{so the gradient error is } g \cdot (1 - \sigma'(x)) \text{ per coordinate.}$$

This mismatch does *not* vanish as $\hat{\sigma} \to \sigma$. It is a property of substituting $\mathbf{I}$ for the true Jacobian of any nonlinearity whose derivative is not identically 1. This is the correct version of the statement; an earlier version of this work asserted that the STE error is bounded by $\epsilon_{\mathrm{ASNC}}/\delta_{\min}$, which conflated the mismatch between $\hat{\sigma}$ and σ (which can be made small by increasing K) with the mismatch between $\mathbf{I}$ and the true Jacobian (which cannot).

We can separate these two effects:

$$\begin{aligned}\text{STE backward minus true backward} &= g \cdot \mathbf{1} - g \cdot \sigma'(x) \\ &= \underbrace{g \cdot (1 - \hat{\sigma}'(x))}_{\text{target-function mismatch}} + \underbrace{g \cdot (\hat{\sigma}'(x) - \sigma'(x))}_{\text{approximation mismatch}}.\end{aligned}$$

The first term is the irreducible cost of using STE at all; its magnitude is $|g| \cdot |1 - \hat{\sigma}'(x)|$. Since $\hat{\sigma}$ is piecewise constant on the grid (deriv. $= 0$ in the interior, undefined at segment boundaries), the first term has magnitude $|g|$ in the interior of segments. The second term vanishes pointwise if $\hat{\sigma}'$ matches σ', which in general requires the codec to be piecewise *affine* rather than piecewise constant; ASNC is piecewise constant, so this term is $|g| \cdot |\sigma'(x)|$, also of order $|g|$.

Proposition E.2 (STE gradient characterization under LASER, restated)**.** *Under Assumptions 4.1–4.13:*

1. ***Linear layers.*** *The STE gradient equals the gradient of the N-bit quantized ANN exactly (Lemma E.1).*
2. ***Nonlinear layers.*** *The STE gradient is $g \cdot \mathbf{1}$ rather than the true $g \cdot \sigma'(x)$. The mismatch has magnitude $|g| \cdot |1 - \sigma'(x)|$, which does not vanish as $\epsilon_{\text{ASNC}} \to 0$. The mismatch has sign-consistent direction with the true gradient wherever $\sigma'(x) > 0$ (which holds globally for monotone nonlinearities like SiLU and row-wise Softmax on their relevant input ranges).*
3. ***Depth behavior.*** *By Corollary D.3, the gradient mismatches at different nonlinear modules do not compound through linear layers in grid units. They combine as an upper-bounded sum across modules, yielding total mismatch $O(M \cdot \max_m |g_m|)$, independent of the number of linear layers between modules.*

Proof. (1) Lemma E.1. (2) By the STE convention and the target/approximation decomposition above. (3) By the same grid-unit non-amplification argument used for the forward pass (Appendix B): gradient mismatches at module m propagate backward through linear layers $A_m^\top$, and under the λ-calibration the grid-unit norm is preserved up to one rounding per gap. □

Remark E.3 (What this result is and is not). Proposition E.2 is a characterization of *where and how* the STE training signal deviates from the true gradient, not a claim that STE training is close to true-gradient training. The practical content is: under LASER, training with STE is equivalent to training an unmodified upstream model with sign-consistent gradient magnitudes of order $|g|$ per nonlinear site, with no compounding across linear layers. Empirically, this suffices for the ASNC calibration objective (200 gradient steps on 1,024 samples) but we do not claim it would suffice for end-to-end SNN training from scratch, which is a harder problem.

F ASNC Formal Specification

F.1 Segmental Codec and BSE Re-encoding

Each segmental codec $\mathcal{C}_i(x)$ uses a spiking neural structure as defined in (6). The decoded analog value $\hat{y}_i$ for each active segment is immediately re-encoded by the BSE encoder to produce $\mathbf{S}_i^{\text{BSE}}(t)$ within the same window T; the overall output follows (7).

F.2 Adaptive Splitting Algorithm

The adaptive splitting procedure and morphology-aware weighting used here are engineering heuristics informed by the Bennett-integral allocation; the specific numerical constants below are tuned on a validation subset and we make no claim that they are uniquely optimal. An ablation over these hyperparameters is not included in this version and is a limitation.

Split criterion.

$$\frac{L_i}{\bar{L}} \cdot \left(1 + \sigma_i^2\right) \cdot \rho_i \;>\; \tau_{\text{adaptive}}, \tag{19}$$

where L_i is the per-segment loss, $\bar{L}$ is the global average loss, σ_i^2 is the loss variance within the segment, and ρ_i is the activation proportion.

Adaptive threshold.

$$\tau_{\text{adaptive}} \;=\; \tau_{\text{base}} \cdot \alpha_{\text{stag}} \cdot \alpha_{\text{prog}}, \tag{20}$$

where τ_{base} is a base threshold and the α factors correct for training stagnation and progress. N_{stag} denotes the number of recent iterations with no loss improvement, N_{active} denotes the current active segment count, L_0 the initial loss, and L_t the current loss.

$$\alpha_{\text{stag}} \;=\; \begin{cases} 0.2, & \text{if } N_{\text{stag}}/N_{\text{active}} > 0.6, \\ 0.4, & \text{if } N_{\text{stag}}/N_{\text{active}} > 0.3, \\ 1.0, & \text{otherwise,} \end{cases} \qquad \alpha_{\text{prog}} \;=\; \begin{cases} 0.4, & \text{if } (L_0 - L_t)/L_0 < 0.005, \\ 0.6, & \text{if } (L_0 - L_t)/L_0 < 0.02, \\ 1.0, & \text{otherwise.} \end{cases} \tag{21}$$

Split point selection. Candidate split points x_s are selected by maximizing

$$S(x_s) = \alpha_1\, E_{\text{sep}}(x_s) + \alpha_2\, C_{\text{cons}}(x_s) + \alpha_3\, B_{\text{bal}}(x_s) + \alpha_4\, Y_{\text{sep}}(x_s), \tag{22}$$

where E_{sep} measures error separation, C_{cons} reflects internal consistency, B_{bal} evaluates data balance, and Y_{sep} captures target separation. The coefficients α_j are hyperparameters; values used in experiments are $(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = (0.4, 0.3, 0.15, 0.15)$.

F.3 Adaptive Weighting Mechanism

Morphology-aware weights.

$$w_i = 0.5\, I_{\text{func}}(i) + 0.35\, I_{\text{error}}(i) + 0.15\, D_i, \tag{23}$$

where $I_{\text{func}}(i)$ is the functional importance of segment i, $I_{\text{error}}(i)$ quantifies its error profile, and D_i is its data density. The coefficients $(0.5, 0.35, 0.15)$ are tuned on a validation split; they are not proven to be optimal.

Regional priority $R_{\text{region}}(i)$.

$$R_{\text{region}}(i) \;=\; \begin{cases} 3.0, & a_i < 0 \text{ and } b_i > 0 \quad \text{(zero-crossing region)}, \\ 2.5, & b_i \le 0 \quad \text{(purely negative region)}, \\ 2.0, & a_i \ge 0 \quad \text{(purely positive region)}, \\ 1.0, & \text{otherwise.} \end{cases} \tag{24}$$

These regional weights concentrate segments in the zero-crossing region of SiLU-like activations, where curvature is highest; this is consistent with the Bennett-integral prescription which allocates intervals in proportion to $(fg)^{1/3}$.